

MIXED TRACES PCB DESIGN COMPETITION

Organized by Mixed Signals, Electronics Association of MEC

VantageF401RE

STM32F401RE Evaluation & Development Board

Final Design and Evaluation Report

Custom Two-Layer STM32F401RET6 Development Platform
Designed in KiCad EDA | Compact 80 mm × 60 mm Form Factor

Status: Completed Design — Gerber / Drill Files Generated for Submission

No physical fabrication was performed; fabrication was not required by the competition.

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1. Executive Summary

VantageF401RE is a custom two-layer development and evaluation board designed around the STM32F401RET6 microcontroller for submission to the Mixed Traces PCB Design Competition, organized by Mixed Signals, the Electronics Association of MEC. The competition specified a fixed microcontroller, a maximum board size of 100 mm × 100 mm, a strict two-copper-layer constraint, and a defined set of mandatory interfaces, while permitting the NUCLEO-F401RE to be consulted as a functional reference rather than a layout template.

Rather than implementing a minimal microcontroller breakout, the design was elected to treat the competition brief as an opportunity to build a compact, general-purpose STM32 development platform that would remain useful beyond the scope of the competition itself. The resulting board integrates a regulated 3.3 V power architecture, a USB-C interface providing both power delivery and USB 2.0 connectivity with dedicated ESD protection, a Serial Wire Debug (SWD) header for external programmer/debugger use, reset and BOOT0 selection circuitry, independently broken-out UART, SPI and I²C interfaces, a 2×10 GPIO expansion header, user and status indicators, an external 8 MHz crystal oscillator, structured decoupling across the digital and analog supply domains, and dedicated test points for laboratory probing.

The complete design was captured and verified in KiCad. All required interfaces were routed within a final board outline of 80 mm × 60 mm, well inside the 100 mm × 100 mm competition limit, using exactly two copper layers. The finished schematic and PCB layout were checked using KiCad's Electrical Rules Check (ERC) and Design Rules Check (DRC) tooling. The completed ERC returned zero violations, and the completed DRC returned zero reported violations, with four intentional exclusions associated with reviewed USB-C connector mechanical hole-clearance checks tied to the selected connector footprint.

Gerber and NC drill files were generated from the finished layout for competition submission. In accordance with the competition rules, no physical fabrication, assembly, or hardware bring-up was performed or is claimed as part of this report; the deliverable is a design-level submission supported by design-level verification. This report documents the problem statement, architecture, circuit-level design decisions, PCB placement and routing strategy, the challenges encountered and how they were addressed, the verification results obtained, and the notable engineering characteristics that distinguish VantageF401RE as a feature-rich, independently developed STM32 evaluation platform.

2. Introduction

2.1 Project Background

STM32-based evaluation boards such as the ST NUCLEO series are widely used as reference platforms for learning embedded systems development, prototyping firmware, and validating peripheral interfacing before committing to a production design. The Mixed Traces PCB Design Competition asked participating teams to design an original two-layer evaluation board around the STM32F401RE microcontroller, using the NUCLEO-F401RE strictly as a functional reference for understanding how the device is typically supported, while prohibiting direct reproduction of its PCB layout.

This constraint framed the project as both a circuit-design exercise and a layout-engineering exercise: the team needed to reproduce the functional guarantees that make an STM32F401RE board usable for development (clean

power, reliable programming access, accessible peripherals) while independently deriving the schematic organization, placement, and routing within a much smaller, self-imposed footprint.

2.2 Problem Statement

The core design problem was to deliver a fully functional, independently laid-out STM32F401RE development board that satisfies a fixed set of mandatory interfaces and protections, fits within a 100 mm × 100 mm competition envelope, and can be routed cleanly using only two copper layers — a layer count that removes the routing flexibility a dedicated internal ground or power plane would otherwise provide.

2.3 Objectives

- Design a complete, reusable STM32F401RET6 development and evaluation platform rather than a minimal MCU breakout.
- Satisfy every mandatory competition interface: MCU, power input, voltage regulation, decoupling, power indication, SWD, reset, user button, user LED, power LED, UART, SPI, I²C, GPIO expansion, and USB.
- Add protective and practical features — USB ESD protection, an external 5 V input path, USB status indication, BOOT0 selection, test points, and dedicated expansion headers — that improve robustness and usability.
- Independently develop the schematic organization, component placement, board dimensions, and routing, using the NUCLEO-F401RE strictly as a functional reference.
- Complete the design to a verifiable state in KiCad and generate fabrication-ready outputs for submission.

2.4 Competition Constraints

The following constraints, summarized in Table 1, governed every design decision made for VantageF401RE.

Table 1. Summary of competition requirements and constraints.

Constraint	Requirement
Microcontroller	Fixed: STM32F401RE / STM32F401RET6
Board type	Original two-layer STM32F401RE evaluation/development board
Maximum board size	100 mm × 100 mm
Copper layers	Exactly two
Reference design	NUCLEO-F401RE permitted as a functional reference only
Layout originality	Direct copying of the NUCLEO-F401RE PCB layout prohibited
Fabrication	Not required for the competition
Mandatory interfaces	MCU, power input, regulation, decoupling, power LED, SWD, reset button, user button, user LED, UART, SPI, I ² C, GPIO expansion, USB

3. Proposed Solution

3.1 Overall Architecture

VantageF401RE is organized around the STM32F401RET6 as the central processing element, with all supporting circuitry grouped into functional subsystems: power input and regulation, USB-C connectivity with ESD protection, programming and debug access, user interaction and indication, clock generation, and peripheral breakout (UART, SPI, I²C, and general-purpose GPIO expansion). Each subsystem was designed, decoupled, and placed as a discrete functional block, which simplified both schematic capture and PCB floor-planning under the two-layer constraint.

Two independent power sources — USB-C 5 V and an externally supplied 5 V input — are combined ahead of a single AP2112K-3.3 linear regulator that produces the board's 3.3 V core rail. This keeps the downstream power distribution, decoupling, and power indication circuitry unified regardless of which supply is in use.

3.2 Functional Block Diagram

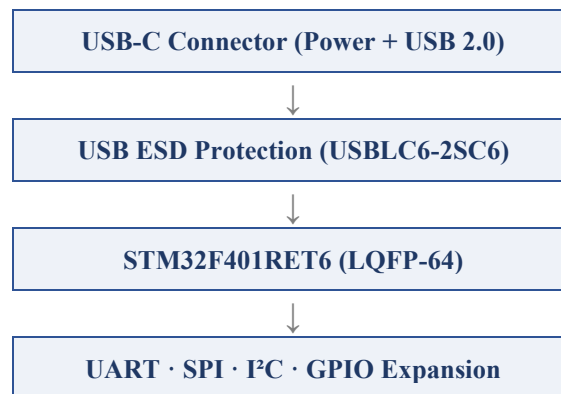


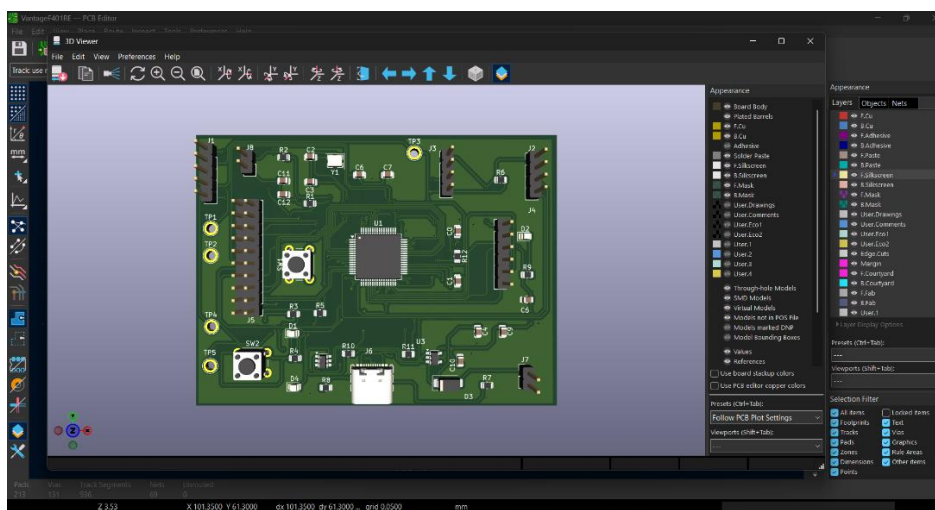
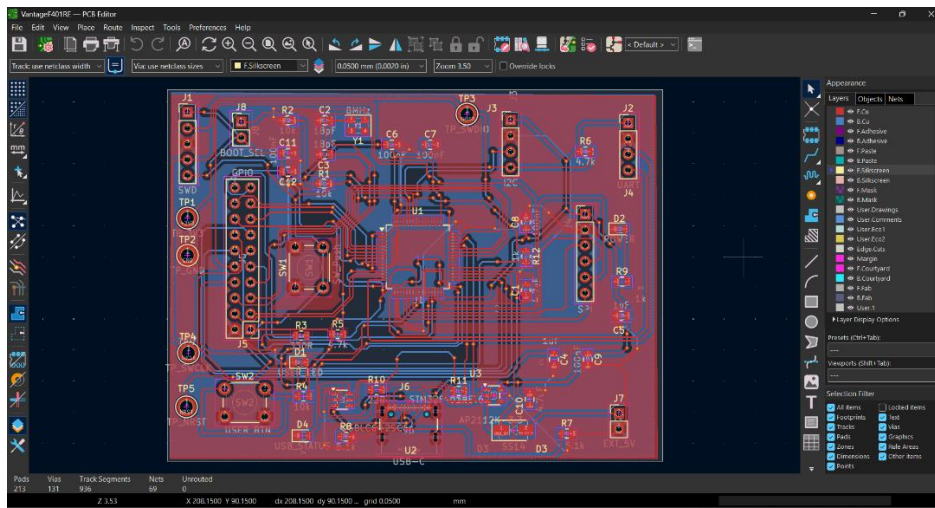
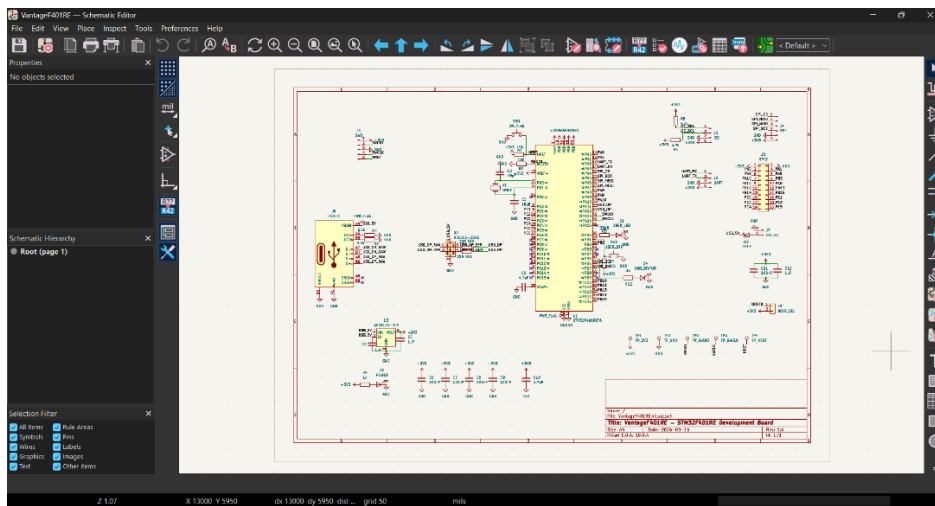
Figure 1. Overall VantageF401RE functional block diagram — primary data and control path.

In parallel with the primary signal path shown in Figure 1, the power subsystem feeds the MCU directly, while SWD, reset, BOOT0, the clock circuit, user controls, indicators, and test points connect to the STM32F401RET6 as supporting subsystems, as summarized in Figure 2.

Supporting Subsystem	Connection to STM32F401RET6
Power (USB-C 5 V / External 5 V via SS14 → AP2112K-3.3 → +3V3)	VDD / VDDA pins (19, 32, 48, 64, 13); VSS/VSSA (18, 31, 47, 63, 12); VBAT (1)
SWD Debug Header (J1)	PA13 (SWDIO), PA14 (SWCLK), NRST
Reset Circuit (R1, SW1)	NRST (pin 7)
BOOT Selection (R2, J8)	BOOT0 (pin 60)
8 MHz Crystal (Y1, C2, C3)	PH0-OSC_IN (pin 5), PH1-OSC_OUT (pin 6)
User LED / Button (D1, SW2)	PB0 (pin 26), PB1 (pin 27)
Test Points (TP1–TP5)	+3V3, GND, SWDIO, SWCLK, NRST

Figure 2. Supporting subsystems connected to the STM32F401RET6.

3.3 Screenshots from KiCad during PCB designing



4. Circuit Architecture

4.1 STM32 Processing Core

The design uses the STM32F401RET6 in the LQFP-64 package (symbol MCU_ST_STM32F4:STM32F401RETx, footprint Package_QFP:LQFP-64_10x10mm_P0.5mm) as the sole processing element. The MCU is placed approximately at the geometric center of the PCB, acting as the primary routing hub around which every functional block is organized.

Power pins are grouped and connected as follows: VDD pins 19, 32, 48 and 64 to +3V3; VSS pins 18, 31, 47 and 63 to GND; the combined VDDA/VREF+ pin 13 to +3V3; the combined VSSA/VREF– pin 12 to GND; and VBAT (pin 1) to +3V3. VCAP1 (pin 30) is decoupled to GND with a dedicated 4.7 μ F capacitor (C1), as required for the internal regulator of the STM32F4 series.

Design Note — Pin Availability

The STM32F401RET6 does not expose PB11 as a usable GPIO in this package/variant; the corresponding physical location is occupied by VCAP1. The GPIO expansion header mapping in this report and in the schematic reflects this correctly and does not present PB11 as an available signal.

4.2 Power System

The board implements a single regulated 3.3 V rail derived from a 5 V input bus (USB_5V) that can be sourced from either the USB-C connector or an external 5 V input. Regulation is performed by U3, an AP2112K-3.3 low-dropout linear regulator in a SOT-23-5 package, wired with VIN (pin 1) and EN (pin 3) to USB_5V, VOUT (pin 5) to +3V3, and GND (pin 2) to GND. Input and output decoupling is provided by C4 (1 μ F, USB_5V to GND) and C5 (1 μ F, +3V3 to GND) respectively, placed close to the regulator body.

The external 5 V input is presented on header J7, where pin 1 passes through D3, an SS14 Schottky diode, before joining the USB_5V net, and pin 2 connects to GND. The SS14 diode provides reverse-current and basic input protection on the externally supplied rail, isolating it from the USB-C supply path while allowing either source to power the board without conflict.

Power presence is indicated by a dedicated LED stage: +3V3 feeds R9 (1 k Ω) in series with D2 (the POWER LED) to GND. Two power test points are also provided — TP1 on +3V3 and TP2 on GND — for convenient meter or oscilloscope probing during laboratory use.

Table 2. Power system component summary.

Reference	Function	Value / Part
U3	3.3 V linear regulator	AP2112K-3.3, SOT-23-5
D3	External-input protection diode	SS14 Schottky
C4	Regulator input decoupling	1 μ F
C5	Regulator output decoupling	1 μ F
R9 / D2	Power indication	1 k Ω / LED

Reference	Function	Value / Part
TP1 / TP2	Power test points	+3V3 / GND

4.3 USB Interface

USB connectivity is provided through J6, a USB-C receptacle (footprint Connector_USB:USB_C_Receptacle_GCT_USB4105-xx-A_16P_TopMnt_Horizontal), supplying both the USB_5V power rail and USB 2.0 full-speed data. Configuration-channel pins CC1 and CC2 are each terminated to GND through 5.1 k Ω resistors, which is the standard passive termination that presents the connector as a USB 2.0 sink/peripheral to an attached USB-C source. The SBU1 and SBU2 sideband pins are left not-connected, as they are not required for a USB 2.0-only implementation.

4.4 USB ESD Protection

The D+ and D– data lines are routed from the USB-C connector into U2, a USBLC6-2SC6 ESD-protection device in a SOT-23-6 package, before continuing to the MCU. The raw connector-side nets USB_DP_RAW and USB_DM_RAW enter U2 at pins 1 and 3 respectively; U2 pin 2 and pin 5 connect to GND and USB_5V for device biasing and clamping reference; and the protected, MCU-side nets USB_DP_SER and USB_DM_SER leave U2 at pins 6 and 4 respectively. From there, 22 Ω series resistors (R10 on D+, R11 on D–) connect to the final USB_DP and USB_DM nets, which terminate at PA12 and PA11 on the MCU.

Design Note — ESD Device Pinout

U2 pin 1 (USB_DP_RAW) and pin 6 (USB_DP_SER) are distinct nets connected through the internal protection element, not shorted together; the same applies to pin 3 (USB_DM_RAW) and pin 4 (USB_DM_SER). This is deliberate: the ESD device is placed electrically between the exposed connector pins and the MCU-facing pins so that transient events at the connector are clamped before reaching PA11/PA12.

Placing U2 close to the USB-C connector, ahead of the series resistors and the MCU, ensures that ESD events are clamped as close as possible to the point of entry, minimizing the length of unprotected trace exposed to the connector.

4.5 SWD Debug Interface

Programming and debugging access is provided through J1, a dedicated 5-pin SWD header, wired as pin 1 = +3V3, pin 2 = SWDIO, pin 3 = GND, pin 4 = SWCLK, and pin 5 = NRST. On the MCU side, SWDIO connects to PA13 (pin 46), SWCLK connects to PA14 (pin 49), and the reset line connects to NRST (pin 7).

VantageF401RE does not include an onboard ST-Link or other integrated debug-probe circuitry. This is a deliberate design choice: omitting the programmer section keeps the board compact and keeps its component budget and layout area focused on the evaluation platform itself, while still providing full programming and debugging capability through any external SWD-compatible probe.

4.6 Reset and BOOT

The reset network consists of R1, a 10 k Ω pull-up from NRST (pin 7) to +3V3, and SW1, a momentary push-button connecting NRST to GND. This provides a debounced-by-design, normally-high reset line that can be asserted manually during development.

BOOT mode selection uses R2, a 10 k Ω pull-down on BOOT0 (pin 60), together with J8, a 2-pin BOOT_SEL header where pin 1 connects to BOOT0 and pin 2 connects to +3V3. With J8 open, BOOT0 is held low by R2 and the device boots from the main flash memory in the normal operating mode; bridging J8 pulls BOOT0 high and selects the system bootloader for in-field reprogramming without an external debug probe. Together, the reset and BOOT circuits give the developer convenient, header-level control over reset and boot-mode configuration during firmware development and recovery.

4.7 UART

J2 is a dedicated 4-pin UART header, wired as pin 1 = +3V3, pin 2 = GND, pin 3 = UART_TX, and pin 4 = UART_RX. On the MCU side, UART_TX connects to PA2 (pin 16) and UART_RX connects to PA3 (pin 17), providing an independent serial interface for logging, host communication, or bootloader interaction separate from the USB path.

4.8 SPI

J4 is a dedicated 6-pin SPI header, wired as pin 1 = +3V3, pin 2 = GND, pin 3 = SPI_SCK, pin 4 = SPI_MISO, pin 5 = SPI_MOSI, and pin 6 = SPI_CS. On the MCU side, SPI_SCK connects to PA5 (pin 21), SPI_MISO to PA6 (pin 22), SPI_MOSI to PA7 (pin 23), and SPI_CS to PA4 (pin 20), giving direct header access to a full SPI master interface for external peripherals such as displays, memory devices, or sensor modules.

4.9 I²C

J3 is a dedicated 4-pin I²C header, wired as pin 1 = +3V3, pin 2 = GND, pin 3 = I2C_SCL, and pin 4 = I2C_SDA. On the MCU side, I2C_SCL connects to PB6 (pin 58) and I2C_SDA connects to PB7 (pin 59). Because I²C is an open-drain bus, R6 and R7 (4.7 k Ω each) provide the pull-up resistors required for SCL and SDA to return to a valid logic-high level; without these pull-ups the bus cannot function, since neither the master nor any slave device actively drives a high level on an open-drain line.

4.10 GPIO Expansion

J5 is a 2 $\times$ 10 general-purpose expansion header exposing ten GPIO signals alongside power and ground, intended to support sensors, displays, external modules, and general prototyping work beyond the fixed peripheral headers. The complete pin mapping is provided in Table 3 and repeated in Appendix B.

Table 3. GPIO expansion header (J5) pin mapping.

Pin	Signal	Pin	Signal
1	+3V3	11	PB12
2	GND	12	PB13
3	PA0	13	PB14

Pin	Signal	Pin	Signal
4	PA1	14	PB15
5	PA8	15	PC0
6	PA9	16	PC1
7	PA10	17	PC2
8	PB2	18	PC3
9	PB10	19	PC4
10	PC6	20	PC5

4.11 User Interface

User interaction is provided by SW2, a push-button on PB1 (pin 27) with a 10 k Ω pull-up to +3V3, and D1, a user-controllable LED on PB0 (pin 26) driven through a 330 Ω series resistor (R3). Status indication is provided separately by D2 (power, described in Section 4.2) and D4 (USB status, described below), keeping power, USB, and user-programmable indication visually distinct on the silkscreen and in use.

4.12 Clock Circuit

An external 8 MHz crystal, Y1 (footprint Crystal_SMD_3225-4Pin_3.2x2.5mm), is connected between PH0-OSC_IN (pin 5) and PH1-OSC_OUT (pin 6), with 18 pF load capacitors C2 and C3 returning each oscillator pin to GND. The crystal and its load capacitors were placed as close as practicable to the MCU oscillator pins, which minimizes stray trace capacitance and loop area around the oscillator network — a standard precaution for reliable crystal startup and frequency accuracy.

4.13 Decoupling

Decoupling is applied across every supply domain on the board rather than as a single bulk capacitor: local 100 nF capacitors (C6–C9) are placed at the MCU's digital VDD pins, a 4.7 μ F bulk capacitor (C10) supports the digital rail against larger transient loads, and the analog supply is separately filtered with C11 (100 nF) and C12 (1 μ F) to reduce coupling of digital switching noise into the ADC/analog reference domain. VCAP1 is decoupled with C1 (4.7 μ F), and the regulator input/output are decoupled locally with C4 and C5 as described in Section 4.2.

This layered approach — small local capacitors for high-frequency transient response placed as close as possible to each supply pin, larger bulk capacitance for lower-frequency energy storage, and a dedicated, isolated analog decoupling path — reflects standard mixed-signal power-integrity practice and helps keep the MCU's internal regulator, digital core, and analog peripherals operating within their supply-noise tolerances.

Table 4. Decoupling and bulk capacitance summary.

Reference	Value	Location / Purpose
C1	4.7 μ F	VCAP1 to GND
C4	1 μ F	Regulator input (USB_5V)

Reference	Value	Location / Purpose
C5	1 μ F	Regulator output (+3V3)
C6–C9	100 nF ($\times 4$)	Local MCU VDD decoupling
C10	4.7 μ F	Bulk decoupling, digital rail
C11	100 nF	Analog supply (VDDA) decoupling
C12	1 μ F	Analog supply filtering

4.14 Test Points

Five labelled test points are included for laboratory probing: TP1 (+3V3), TP2 (GND), TP3 (SWDIO), TP4 (SWCLK), and TP5 (NRST). These provide convenient, connector-free access to the primary power and debug-critical nets. No physical measurements have been taken at these points as part of this submission; their purpose is to support future bring-up and debugging once hardware is available.

5. PCB Design

5.1 Board Dimensions

The finished PCB measures 80 mm × 60 mm, well within the 100 mm × 100 mm maximum permitted by the competition. This compact envelope was chosen deliberately, rather than filling the full allowed area, to demonstrate that the complete required and supplementary feature set could be achieved in a genuinely compact form factor suitable for bench use, enclosure integration, or embedding in a larger prototype.

5.2 Layer Stack / Two-Layer Constraint

The board uses exactly two copper layers (F.Cu and B.Cu), as required by the competition. Working within two layers removes the option of a dedicated internal ground plane or internal power plane, so return-current paths and power distribution had to be managed directly on the top and bottom copper using ground fill zones and deliberate routing discipline rather than being handled implicitly by an inner plane.

5.3 Component Placement

The STM32F401RET6 was placed centrally on the board, consistent with its role as the primary routing hub described in Section 4.1. Functional blocks were arranged around it by proximity to their associated MCU pins and by their external connection requirements:

- The USB-C connector (J6) is positioned at the board edge to allow a cable to be inserted without obstruction, with the USB ESD protection device (U2) placed close to it so that the unprotected trace run from connector to ESD device is minimized.
- The power regulator (U3) is located near the USB/power entry point, with its input and output decoupling capacitors (C4, C5) placed immediately adjacent to its pins.
- The crystal (Y1) and its load capacitors (C2, C3) are placed close to the MCU oscillator pins (PH0/PH1) to keep the oscillator loop short.
- The SWD header (J1) is kept accessible at a board edge for probe or cable connection.
- UART, I²C, and SPI headers (J2, J3, J4) are grouped together to simplify wiring during bench use.
- The GPIO expansion header (J5) is positioned for unobstructed access along its own edge.
- The reset button, user button, and all four indicator LEDs are placed for physical accessibility and clear visibility respectively.
- Test points (TP1–TP5) are distributed near their associated nets for convenient probing without interfering with other components.

This placement strategy reduces unnecessary trace length between related components, keeps high-priority short connections (decoupling, oscillator, ESD protection) physically short by construction, and groups connectors and controls logically for a developer using the board on a bench.

5.4 Routing Strategy

Routing followed the functional grouping established during placement: signals were kept within their functional region wherever possible, power connections were kept short and direct, and the USB differential pair was routed as a matched, closely-spaced pair from the connector through the ESD device and series resistors to the MCU, without claiming formal impedance-controlled routing, since impedance control was not implemented or documented as part of this submission.

Routing a feature set of this breadth — USB, dual power inputs, SWD, UART, SPI, I²C, a ten-line GPIO expansion header, four indicator LEDs, two user controls, and five test points — using only two copper layers required continuous trade-offs between direct routing and layer changes. Ground-fill zones were used on both layers to recover return-path continuity that a two-layer board cannot obtain from an internal plane, and via placement for layer changes was kept close to the pins being connected to minimize additional trace length introduced by each transition.

5.5 Ground Planes

Copper-fill ground zones were poured on both the top and bottom layers, stitched together where practical, rather than relying solely on discrete ground traces. On a two-layer board, these zones serve a genuinely practical role: they provide return-current paths for signals without requiring an explicit routed ground trace alongside every signal, they reduce the amount of dedicated ground routing needed elsewhere, and they generally improve the organization and manufacturability of the copper. These fills are described here as a practical layout aid consistent with standard two-layer design practice; no claim of guaranteed EMI compliance or formal electromagnetic-compatibility qualification is made for this design.

5.6 Power Distribution

The 3.3 V rail is distributed from the regulator output outward to the MCU's four VDD pins and the VDDA/VBAT pins, with local decoupling at each destination as described in Section 4.13. Keeping the regulator physically close to the MCU, and keeping each decoupling capacitor physically close to its associated pin, was prioritized over routing convenience wherever the two conflicted, consistent with standard power-integrity practice for MCU-class digital devices.

5.7 USB Layout Considerations

The USB-C connector, ESD protection device, and series resistors were placed and routed as a contiguous group at the board edge, with the differential D+/D− pair kept short and routed together from the connector, through U2, through R10/R11, and into PA12/PA11. As noted in Section 5.4, this reflects careful physical grouping and short, matched-length routing rather than a formally impedance-controlled transmission-line design, which was not implemented or documented for this board.

6. Major Components

Table 5 summarizes the principal active and passive components used in the design, beyond the general-purpose resistors and capacitors already itemized in Sections 4.2 and 4.13.

Table 5. Major components and their roles.

Reference	Component	Package / Footprint	Role
U1	STM32F401RET6	LQFP-64, 10×10 mm, 0.5 mm pitch	Main microcontroller
U2	USBLC6-2SC6	SOT-23-6	USB D+/D− ESD protection
U3	AP2112K-3.3	SOT-23-5	3.3 V LDO regulator
Y1	8 MHz crystal	SMD 3225-4Pin, 3.2×2.5 mm	System clock reference
J6	USB-C receptacle	GCT USB4105-xx-A, 16-pin	Power input + USB 2.0 data
D3	SS14 Schottky diode	SMA/SMB (discrete)	External 5 V input protection
J1	5-pin header	2.54 mm pitch	SWD debug interface
J2 / J3 / J4	4- / 4- / 6-pin headers	2.54 mm pitch	UART / I ² C / SPI breakout
J5	2×10 header	2.54 mm pitch	GPIO expansion
J7	2-pin header	2.54 mm pitch	External 5 V input
J8	2-pin header	2.54 mm pitch	BOOT0 selection
D1 / D2 / D4	LEDs	SMD	User / power / USB status indication
SW1 / SW2	Tactile push-buttons	SMD	Reset / user button

7. Major Design Decisions

Building a complete platform, not a minimal breakout

I chose to implement the full set of practical interfaces a developer would expect from an evaluation board — independently accessible UART, SPI, and I²C, a general-purpose GPIO expansion header, and both status and user-programmable indicators — rather than the minimum required to satisfy the competition checklist. This decision shaped nearly every subsequent layout and component choice, since it committed the design to fitting a broader feature set into the same two-layer, 100 mm × 100 mm envelope.

Dual power input with shared regulation

Supporting both USB-C and an external 5 V input through a single shared regulator and decoupling path (Section 4.2) was chosen over implementing two independent regulation paths, in order to keep the power distribution and indication circuitry unified and to avoid duplicating decoupling and test-point infrastructure for two rails that ultimately serve the same purpose.

Dedicated ESD protection ahead of the MCU

Rather than connecting the USB-C data lines directly to the MCU through only series resistors, I added the USBLC6-2SC6 ESD-protection device specifically to intercept transient events at the connector boundary, reflecting the exposed nature of a user-facing USB-C port on a development board that will be repeatedly connected and disconnected.

External SWD rather than an onboard programmer

Integrating an onboard ST-Link-class programmer was considered and deliberately not pursued, in favor of a dedicated external SWD header. This kept the component count, board area, and schematic complexity focused on the evaluation platform itself, while still providing complete programming and debugging capability through widely available external probes.

Independent peripheral headers over a single combined connector

UART, SPI, and I²C were each broken out to their own dedicated header rather than combined onto a single generic connector, so that each interface could be connected to external hardware unambiguously and without adapter cabling, at the cost of additional header footprint on the board.

8. Design Considerations

Beyond the major decisions above, the following practical considerations informed component and feature selection throughout the design:

- Practical usefulness — each added feature (external 5 V input, USB status LED, BOOT0 header, test points) was evaluated for genuine development-time value rather than being included simply to lengthen a feature list.
- Board area — every header and component had to justify its footprint within the self-imposed 80 mm × 60 mm outline, which is well under the 100 mm × 100 mm ceiling.
- Component count — passive count was kept proportionate to function; for example, decoupling was placed per supply domain rather than as a single oversized bulk capacitor, but no redundant capacitors were added beyond what each domain required.
- Cost — commodity, widely available parts (SOT-23-class regulators and protection devices, standard 0.1 inch headers) were favored over specialized components where function was equivalent.
- Routing complexity — features that would have required significant additional routing area or layer changes without proportional functional benefit were avoided, consistent with the two-layer constraint discussed in Section 5.2.
- Development value — interfaces likely to be used repeatedly during firmware development (UART, SWD, GPIO expansion) were prioritized for dedicated, unambiguous header access.
- Reliability and protection — the external-input protection diode and the USB ESD device were included specifically to improve the board's tolerance of real bench-use conditions such as hot-plugging and miswired external supplies.

The design deliberately avoided adding features purely to increase apparent complexity; every block described in Section 4 maps to a specific competition requirement or a clearly stated practical benefit.

9. Challenges and Solutions

A. Two-layer limitation

Challenge: Many interfaces and support circuits had to be accommodated within only two copper layers, without the routing flexibility an internal plane would provide.

Solution: Functional partitioning of the schematic, central MCU placement, perimeter-oriented header placement, disciplined use of ground-fill zones, and iterative routing passes to resolve congestion.

Outcome: A completed two-layer PCB layout satisfying every required interface within the target footprint.

B. Feature density

Challenge: USB, dual-source power, SWD, UART, SPI, I²C, GPIO expansion, four indicators, two user controls, and five test points all needed to coexist on an 80 mm × 60 mm board.

Solution: Consistent functional grouping during placement (Section 5.3) and deliberate, unambiguous connector choice for each interface.

Outcome: All required and supplementary features fit within the target footprint without compromising accessibility.

C. USB integration

Challenge: The USB-C connector had to simultaneously provide 5 V power delivery and USB 2.0 data connectivity while incorporating ESD protection ahead of the MCU.

Solution: Adoption of the USBLC6-2SC6 protection device, correctly configured CC-termination resistors, 22 Ω data-line series resistors, and structured, unambiguous net naming (USB_DP_RAW / USB_DP_SER / USB_DP, and the corresponding D[−] nets) to keep the signal path traceable through the schematic.

Outcome: A protected, correctly referenced USB 2.0 data and power path from connector to MCU.

D. Multiple power inputs

Challenge: USB-C and the external 5 V input both needed to feed the same regulated power system without one source interfering with or being back-driven by the other.

Solution: Routing the external 5 V input through D3 (SS14) before it joins the shared USB_5V net, isolating the external path while allowing either source to supply the regulator.

Outcome: A single, unified 5 V bus feeding one regulation and decoupling path regardless of source.

E. MCU power integrity

Challenge: The STM32F401RET6 presents multiple VDD/VSS pin pairs, a VCAP1 pin, and a separate analog supply domain, each with its own decoupling requirement.

Solution: Local 100 nF capacitors at each digital VDD pin, a 4.7 μF bulk capacitor for the digital rail, dedicated analog-domain capacitors (C11, C12), and a VCAP1 capacitor (C1) sized and placed per ST's device guidance.

Outcome: A fully decoupled power delivery network across all MCU supply domains, as documented in Section 4.13.

F. Originality requirement

Challenge: The NUCLEO-F401RE could be consulted as a functional reference, but direct reproduction of its PCB layout was explicitly prohibited by the competition.

Solution: Independent development of the schematic organization, component placement, board outline dimensions, and routing, using the NUCLEO-F401RE only to confirm functional expectations such as pin usage and typical supporting circuitry.

Outcome: An independently laid-out board, discussed further in Section 19 (Originality).

G. Final verification

Challenge: The completed schematic and PCB needed to be confirmed compliant with KiCad's rule-checking tools before the design could be considered finished.

Solution: Iterative ERC and DRC checking, with each reported issue reviewed and either corrected in the design or, where appropriate, formally excluded after engineering review.

Outcome: A clean final verification result, detailed in Section 10.

10. Verification and Validation

10.1 Electrical Rules Check (ERC)

The completed schematic was checked using KiCad's built-in Electrical Rules Check. The final run returned zero violations and zero warnings, confirming that pin-type conflicts, unconnected required pins, and other schematic-level electrical rule issues had been resolved prior to submission.

Table 6. Final ERC result.

Metric	Result
Violations	0
Errors	0
Warnings	0

10.2 Design Rules Check (DRC)

The completed PCB layout was checked using KiCad's Design Rules Check. The final run reported zero DRC violations. Four items were intentionally excluded after engineering review; these correspond to mechanical hole-clearance checks associated with the selected USB-C connector footprint (GCT USB4105-xx-A), which is a known and reviewed characteristic of that footprint rather than an unresolved layout defect.

Table 7. Final DRC result.

Metric	Result
Reported violations	0
Errors	0
Intentional exclusions	4 (reviewed USB-C connector mechanical hole-clearance checks)

Verification Statement

0 reported DRC violations, with 4 reviewed intentional exclusions. These exclusions were examined individually and confirmed to relate to the mechanical hole-clearance geometry of the selected USB-C connector footprint, not to an unresolved electrical or manufacturability issue.

10.3 Design-Level Verification

Beyond the automated ERC/DRC checks, the design was reviewed at the schematic level for correct net connectivity against the MCU pin mapping documented in Appendix A, correct polarity and orientation of the ESD-protection and regulator devices, and correct pull-up/pull-down placement on the reset, BOOT0, and I²C nets. This review was performed within KiCad and does not constitute physical electrical measurement; no oscilloscope, multimeter, or other bench instrument was used, since no physical board exists at this stage of the submission.

10.4 Fabrication Outputs

Gerber and NC drill files were generated from the finished PCB layout for competition submission. The exported set includes the standard manufacturing layers: F.Cu, B.Cu, F.Mask, B.Mask, F.Silkscreen, B.Silkscreen, F.Paste, B.Paste, Edge.Cuts, the associated drill files, and a Gerber job file where applicable.

Fabrication Status

Gerber and drill files were generated for fabrication/submission. No physical fabrication, assembly, or hardware bring-up has been performed or is claimed for VantageF401RE as part of this report; fabrication was not required by the competition.

11. Distinguishing Design Features

The following characteristics, summarized in Table 8, are presented as notable engineering features of the VantageF401RE design that can be independently verified from the submitted schematic and PCB layout. These features provide a strong and technically complete feature set for an STM32 development platform and represent areas in which the design demonstrates additional engineering consideration; they are not presented as a claim of superiority over any other team's submission.

#	Notable Design Characteristic
1	Complete STM32 development platform rather than a minimal breakout
2	USB-C connector providing both 5 V power and USB 2.0 connectivity
3	Dedicated USB ESD protection (USBLC6-2SC6) ahead of the MCU
4	22 Ω series resistors on both USB data lines
5	External 5 V input option in addition to USB-C power
6	SS14 Schottky protection on the external 5 V input path
7	Dedicated 5-pin SWD header for external programmer/debugger use
8	BOOT0 selection header for convenient bootloader entry
9	Dedicated reset button and independent user button
10	UART, SPI and I ² C each exposed on independent, dedicated headers
11	2×10 GPIO expansion header for sensors, displays and prototyping
12	Five dedicated test points on power and debug-critical nets
13	Separate power-indication and USB-status LEDs
14	External 8 MHz crystal placed close to the MCU oscillator pins
15	Structured decoupling across digital, analog, and VCAP supply domains
16	Ground-fill copper zones on both layers
17	Compact 80 mm × 60 mm form factor, well inside the 100 mm × 100 mm limit
18	Full required feature set achieved within a two-layer stack-up
19	Completed schematic and PCB verified via ERC/DRC prior to submission
20	Gerber and drill files generated and packaged for submission

12. Final Outcome

The VantageF401RE design was carried through to a complete, verified state satisfying every mandatory competition requirement listed in Table 1, while adding the supplementary features described in Section 11. The finished layout occupies 80 mm × 60 mm on a two-layer PCB, passed KiCad's ERC with zero violations, passed KiCad's DRC with zero reported violations and four reviewed intentional exclusions, and produced a complete Gerber and drill package for submission.

Table 9. Final design specification summary.

Parameter	Specification
Project	VantageF401RE
MCU	STM32F401RET6, LQFP-64
PCB	2-layer
Board dimensions	80 mm × 60 mm
Power input	USB-C 5 V / External 5 V (J7)
Regulator	AP2112K-3.3
Core rail	3.3 V
USB	USB-C connector, USB 2.0
USB protection	USBLC6-2SC6
Debug interface	SWD (external probe)
Clock	8 MHz external crystal
Serial interfaces	UART, SPI, I ² C (dedicated headers)
Expansion	2×10 GPIO header
User controls	Reset button, user button, BOOT0 selection header
Indicators	User LED, Power LED, USB status LED
Test points	+3V3, GND, SWDIO, SWCLK, NRST
ERC	0 violations
DRC	0 reported violations, 4 reviewed intentional exclusions
Fabrication status	Completed design — Gerber/drill files generated — no fabrication claimed

13. Limitations and Future Work

VantageF401RE is submitted as a completed design verified at the schematic and PCB-layout level. As fabrication was not required by the competition and was not performed, the following boundaries apply to the current submission. These are stated as the natural boundary between design-level verification and physical prototype validation, not as shortcomings of the design work performed:

- No physical fabrication was performed, as fabrication was not required by the competition.
- Physical electrical measurements (voltage, current, ripple, timing) have not been performed, since no assembled board exists.
- USB hardware enumeration has not been physically verified on a host system.
- Regulator thermal performance under load has not been physically measured.
- Signal integrity, including the USB differential pair, has been verified at the design and layout level only, not experimentally with a network analyzer or TDR.

Future work, should the design proceed beyond the competition, would naturally include fabrication and assembly, hardware bring-up and power-on testing, firmware development and validation, USB enumeration testing against a host operating system, oscilloscope-based signal and power-rail measurements, thermal measurement of the regulator under representative load, and further PCB optimization informed by findings from physical testing.

14. Conclusion

VantageF401RE demonstrates that the mandatory requirements of the Mixed Traces PCB Design Competition can be met, and substantially extended, within a genuinely compact two-layer board. Starting from a fixed STM32F401RE microcontroller and a strict set of constraints on size, layer count, and layout originality, I built a complete evaluation and development platform incorporating dual-source power, protected USB-C connectivity, external debug access, independent UART/SPI/I²C breakout, GPIO expansion, user controls and indicators, and structured decoupling, and carried it through to a design verified clean by KiCad's ERC and DRC tools.

The NUCLEO-F401RE was consulted only as a functional reference; the schematic organization, component placement, board dimensions, and routing presented in this report I independently developed. With Gerber and drill files generated and packaged for submission, VantageF401RE is presented as a compact, feature-rich, independently designed STM32F401RE development platform with documented design-level verification, intended to remain useful as a development tool beyond the scope of the competition itself.

Appendix A — MCU Pin Mapping

This appendix consolidates the complete STM32F401RET6 pin usage referenced throughout the report.

Table 10. Power and ground pin mapping.

Signal	Pin(s)	Net
VDD	19, 32, 48, 64	+3V3
VSS	18, 31, 47, 63	GND
VDDA / VREF+	13	+3V3
VSSA / VREF–	12	GND
VBAT	1	+3V3
VCAP1	30	4.7 μ F to GND (C1)

Table 11. Clock, reset, and BOOT pin mapping.

Signal	Pin	Net / Function
PH0-OSC_IN	5	8 MHz crystal (Y1)
PH1-OSC_OUT	6	8 MHz crystal (Y1)
NRST	7	Reset network (R1, SW1) / SWD / J1 / TP5
BOOT0	60	R2 pull-down, J8 BOOT_SEL

Table 12. Interface and peripheral pin mapping.

Signal	Pin	Net / Function
PA13	46	SWDIO (J1)
PA14	49	SWCLK (J1)
PB0	26	USER_LED (D1) via R3
PB1	27	User button (SW2), 10 k Ω pull-up
PA2	16	UART_TX (J2)
PA3	17	UART_RX (J2)
PB6	58	I2C_SCL (J3), R6 pull-up
PB7	59	I2C_SDA (J3), R7 pull-up
PA5	21	SPI_SCK (J4)
PA6	22	SPI_MISO (J4)
PA7	23	SPI_MOSI (J4)
PA4	20	SPI_CS (J4)

Signal	Pin	Net / Function
PA11	44	USB_DM
PA12	45	USB_DP
PB5	57	USB_STATUS (D4) via R12

Note

PB11 is not available as a GPIO on the STM32F401RET6; this physical location is occupied by VCAP1 (pin 30) and is not present in the GPIO expansion mapping in Appendix B.

Appendix B — Interface Header Mapping

Table 13. J1 — SWD header (5-pin).

Pin	Signal
1	+3V3
2	SWDIO
3	GND
4	SWCLK
5	NRST

Table 14. J2 — UART header (4-pin).

Pin	Signal
1	+3V3
2	GND
3	UART_TX
4	UART_RX

Table 15. J3 — I²C header (4-pin).

Pin	Signal
1	+3V3
2	GND
3	I2C_SCL
4	I2C_SDA

Table 16. J4 — SPI header (6-pin).

Pin	Signal
1	+3V3
2	GND
3	SPI_SCK
4	SPI_MISO
5	SPI_MOSI
6	SPI_CS

Table 17. J5 — GPIO expansion header (2×10).

Pin	Signal	Pin	Signal
1	+3V3	11	PB12
2	GND	12	PB13
3	PA0	13	PB14
4	PA1	14	PB15
5	PA8	15	PC0
6	PA9	16	PC1
7	PA10	17	PC2
8	PB2	18	PC3
9	PB10	19	PC4
10	PC6	20	PC5

Table 18. J7 — External 5 V input (2-pin) and J8 — BOOT_SEL (2-pin).

Header	Pin	Signal
J7	1	5V_EXT → D3 (SS14) → USB_5V
J7	2	GND
J8	1	BOOT0
J8	2	+3V3

Appendix C — Repository Structure

The supporting design files for VantageF401RE are organized in the project's GitHub repository as summarized below, separating the editable KiCad source, generated verification reports, fabrication outputs, documentation, and the bill of materials into clearly labelled directories.

```
VantageF401RE/
├── bom/
│   └── Final Component Selection.xlsx
├── docs/
│   ├── Major Decisions.md
│   └── Vantage Initial Block Diagram.png
├── kicad/
│   ├── README.md
│   ├── VantageF401RE.kicad_pro
│   ├── VantageF401RE.kicad_sch
│   └── VantageF401RE.kicad_pcb
├── results/
│   ├── README.md
│   ├── VantageF401RE_ERC_Report.html
│   └── VantageF401RE_DRC_Report.html
├── gerbers/
│   └── VantageF401RE_Gerbers.zip
├── pdf/
│   └── VantageF401RE_Final_Design_Report.pdf
└── README.md
```